

Context-aware vision-language model agent enriched with domain-specific ontology for construction site safety monitoring

Chak-Fu Chan^a, Peter Kok-Yiu Wong^{a,*}, Xiaowen Guo^a, Jack C.P. Cheng^{a,b,*},
Jolly Pui-Ching Chan^c, Pak-Him Leung^b, Xingyu Tao^a

^a Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong Special Administrative Region

^b AutoSafe Limited, Hong Kong Special Administrative Region

^c Drainage Services Department, The Government of the Hong Kong Special Administrative Region, Wanchai, Hong Kong Island, Hong Kong Special Administrative Region

ARTICLE INFO

Keywords:

Construction safety ontology
Construction site safety monitoring
Domain-tailored prompt engineering
Context-aware vision-language model
Virtual construction safety assistant

ABSTRACT

Traditional approaches of construction site safety monitoring heavily rely on manual on-site inspection, which are prone to overlooked incidents. Existing computer vision methods require time-consuming and case-by-case data labeling, and lack high-level reasoning capability. This paper develops a human-like virtual assistant agent by integrating a multi-modal vision-language model into video analytics: (1) To efficiently generate image-text data for model development, a semi-automatic image-text labeling pipeline based on in-context learning is designed; (2) To optimize a virtual agent from pre-trained to domain-tailored, a two-stage curriculum learning paradigm is designed to enhance model fine-tuning effectiveness toward domain-specific tasks; (3) To inject construction-domain knowledge more effectively into the virtual agent, a hierarchical prompting framework driven by a construction safety ontology is developed for more domain-tailored reasoning capability. The virtual agent has been deployed on a real construction site for real-time video analytics, with over 90 % accuracy in identifying violations of work-at-height safety regulations.

1. Introduction

The construction sector is recognized as one of the most dangerous worldwide, with a history of high accident and casualty rates. From 2013 to 2019, the construction sector in Hong Kong has consistently recorded the highest death rates compared to the other 14 primary industry sectors [1,2]. Similarly, in 2023, the United States reported that approximately 20 % of work-related deaths took place at construction sites [3]. Many of the casualties are related to non-compliance with construction safety rules. Construction sites are characterized by safety risks that arise from complex interactions between numerous workers and machinery. To prevent safety risks and serious injuries, it's crucial to detect and analyze any unsafe practices among workers in real time. This can involve monitoring workers' activity on-site and ensuring they're complying with the safety rules. Traditional site safety monitoring mainly involves regular on-site safety inspection by management personnels and safety inspectors, which is labor-intensive and easily leads to overlooking in unsafe behavior. In recent years, computer vision

(CV) techniques based on convolutional neural networks (CNN) are widely adopted for identifying unsafe worker behaviors [4,5], checking personal protective equipment (PPE) compliance [6] and analyzing heavy machinery operation [7–9].

However, CNN-based image/video analytics has several limitations in practical applications. First, developed CV models are specialist models only well-trained for a small subset of detection tasks, thus limited by the narrow domain of knowledge. Case-specific mathematical formulation based on the bounding boxes' size and coordinates are needed for analyzing for particular safety rules, as illustrated by Khan et al. [10] using CNN to analyze worker-scaffold relationship, which may require a sophisticated system design and much human effort. Second, the architecture of CNN-based models leads to limited embedded semantic information (e.g. object category and localization), while extracting semantic information (e.g., construction activities, interactions between different construction objects) from construction images is an essential step for further CV-based application in construction management [11,12]. Importantly, existing CV models still

* Corresponding author at: Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong Special Administrative Region.

E-mail address: cejcheng@ust.hk (J.C.P. Cheng).

<https://doi.org/10.1016/j.autcon.2025.106305>

Received 6 December 2024; Received in revised form 25 May 2025; Accepted 26 May 2025

Available online 2 June 2025

0926-5805/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

lack in-depth reasoning capability to perform semantic understanding, while construction applications like safety compliance checking require high-level reasoning of worker behaviors and on-site conditions. While combining the CV models with an external ontology-based module (i.e., by exporting the outputs of a specialist model to a carefully designed rule-checking system) can acquire a higher level of semantic reasoning, the process is time-consuming and requires constant model or system updates when adapting to various site conditions. More importantly, it is not an end-to-end process, which means any information lost in the vision modality will not be transferred to the external module and represented in the language modality.

To address the limitation of traditional CNN-based or ontology-based methods in processing multi-modal semantic information, a more comprehensive and end-to-end method should be considered. Recently, multi-modal large language models (MLLM) are widely studied upon their multi-modal conversational capability, upon which vision-language models (VLM) combine image processing and natural language processing (NLP). For example, GPT-4 V [13] and Grounding-DINO [14] can extract both intricate visual and linguistic features, undergo deep feature fusion and generate detailed image descriptions [15], perform multiple types of instruction-following tasks such as image captioning, visual grounding and visual question answering (VQA) [16]. Toward construction applications, construction safety rules in textual format can potentially be extracted from safety handbooks or guidelines [17], given to the VLM as queries to obtain intuitive answers in natural language, such as detailed incident descriptions with recommended actions reported to site engineers/safety supervisors [18]. VLMs show potential for human-like cross-modal reasoning ability in construction safety, with the image, instruction and embedded safety rules inputs undergoing deep fusion in an end-to-end approach for more efficient and effective data processing and a higher level of semantic understanding. However, there remain several essential challenges when trying to adapt generic VLMs for construction safety monitoring:

- 1) *Lacked multi-modal data for developing domain-tailored VLMs:* A huge volume of labeled, high-quality and multi-modal data are required for domain adaptation. The construction industry suffers from limited amount of structured data for AI model training, and currently lacks public datasets for effectively fine-tuning VLMs toward diverse construction safety tasks. Existing foundation and open-source VLMs, such as CogVLM [19] and LLaVA [20], require a vast amount of images for comprehensive pre-training or even for supervised fine-tuning, for which manual annotation is labor-intensive. Hence, an efficient methodology for image-text data labeling will greatly facilitate the development of domain-tailored VLMs.
- 2) *Lacked an efficient fine-tuning strategy for developing domain-tailored VLMs:* Similar to CNN-based models, VLMs are mainly pre-trained on internet crowdsourced images, which have notably different image quality as those captured on real construction sites by surveillance cameras. There is a significant data distribution shift when trying to adapt a VLM for real-site downstream tasks, while the industry lacks open-access real-site data for fine-tuning. Moreover, the fine-tuning of VLMs on construction safety domain is not only dependent on the images, but also dependent on the high-quality instruction-response sets for learning domain-specific knowledge, which are also not publicly available and require intensive annotation. Efficient learning of domain knowledge with limited and imbalanced vision-text data tailored for safety monitoring remains a critical research gap that needs to be addressed by special fine-tuning strategies.
- 3) *Lacked integration of construction domain knowledge into VLMs:* Existing AI-based solutions primarily performed object detection and classification from site images, while lacking a standardized representation of domain-specific knowledge crucial for accurate safety monitoring. Although VLMs are a potential solution, they are

originally pre-trained with diverse data which only capture generic knowledge on object recognition and text understanding, while often fall short in logical reasoning within construction safety contexts. Although fine-tuning on domain datasets can already enhance the general ability of the VLM in construction domain, in-context prompting can further adapt it to diverse site-specific tasks effectively and improve its reasoning and compliance checking in site safety. To integrate structured safety knowledge into AI inference process, a construction safety ontology is developed and embedded into a domain-tailored prompting framework to enable knowledge-enhanced reasoning of VLMs.

To address these essential challenges in adapting VLMs for construction safety monitoring, we develop a holistic framework to tackle the problems, leading to the following specific research objectives to be achieved in this paper:

- 1) *A semi-automatic data generation pipeline to address the data availability problem:* We develop a streamlined workflow to generate high-quality, multi-modal instruction-following dataset for aligning the VLM toward construction safety compliance monitoring, greatly reducing the time required to prepare such a comprehensive dataset.
- 2) *A two-stage curriculum learning paradigm for fine-tuning VLM to address the data imbalance and efficiency problem:* For the first stage of the training, we mainly include crowd-sourced images with some descriptive tasks assigned, as these data are more available and can quickly adapt the VLM from a pre-trained model to a construction context-aware model. For the second stage, we incorporate the less available real-site data with site-specific instruction-following tasks, to further fine-tune the task-specific safety monitoring ability of the VLM.
- 3) *Design of a generic prompting framework driven by a construction safety ontology to address the domain knowledge problem:* We embed the construction-related knowledge from an established ontology into a hierarchical prompting framework of the VLM, which can further enhance the object localization and rule analysis capabilities of the model in an effective end-to-end manner. The framework can also adapt to diverse construction environments easily by changing the prompts in natural language.

After development of the VLM based on the above objectives, we further integrate it with real-time video analytical system for on-site testing and evaluation. Safety officers can easily interact with the designed software for automatic safety alarming and comprehensive question answering, greatly reducing human monitoring efforts when compared to conventional methods.

2. Literature review

To help with developing a VLM and ontology based safety monitoring system which accomplishes the research objectives, we have reviewed the current application and limitations of VLM, prompt engineering and ontology methods in construction safety analysis.

2.1. Vision-language models for construction safety analysis

Current VLMs are mainly based on the Vision Transformer-Linear layer-LLM architecture, which treats an image as a sequence of visual tokens from flattened patches, similar to how a language model treats a sentence as a sequence of word tokens. Several methods have been developed for multi-modal feature alignment, such as shallow alignment via a trainable linear layer for embedding mapping [21] or image-text joint training adopted by PaLI [22] and Qwen-VL [23]. To develop domain-tailored VLMs, those foundation models pre-trained on general tasks can be fine-tuned to suit various specific downstream tasks. This is achieved by continuing the training process on more human-supervised

task-specific datasets.

Fan et al. [24] fine-tuned the MiniGPT-4 with a hand-crafted dataset specifically for identifying ergonomic risks of workers. However, they reported hallucination problems among their evaluation, possibly due to the limited volume and diversity of their training data (for such a specific task). Jung et al. [25] generated daily construction reports via keyword-based image retrieval to extract relevant site video footages, with a vision-language transformer to describe the site conditions and activities on each image. However, such captions only describe what happens on a site, rather than performing practical safety compliance checking which requires more high-level reasoning.

Traditionally, for pre-training of LLM, a lot of data and computational resources are required due to the scaled-up model parameters. A more effective fine-tuning technique is necessary to ensure a smoother learning process on various downstream tasks. Parameter-Efficient Fine-Tuning (PEFT) is always used to adapt large pre-trained models to downstream tasks by only fine-tuning a small number of model parameters or additional small-size adapters [26]. This approach significantly decreases the computational and storage costs required to finetune LLMs. Low-Rank Adaptation (LoRA) is a specific method of PEFT that performs on-par or even better than full fine-tuning in model quality when fine-tuning on task-specific datasets, despite having significantly fewer trainable parameters [27]. LoRA freezes the pre-trained model weights and injects trainable low-rank decomposition matrices into selected layers of the Transformer architecture.

2.2. Prompt engineering for construction safety analysis

Prompt engineering is a technique that incorporates task-specific hints or instructions to enable LLMs to adapt to new tasks without supervised learning processes. Different prompting strategies include task instruction prompting [28], in-context learning [29], retrieval-based prompting [30], and chain-of-thought prompting [31].

Several studies leveraged prompt engineering to adapt pre-trained VLMs toward construction applications. Yong et al. [32] examined the effectiveness of prompts in improving zero-shot and few-shot classification and detection of building defects. Wong et al. [33] designed a two-step prompting strategy for identifying project risks from contract clauses, based on in-context learning and majority voting mechanism; yet they only adopted a pre-trained LLM for project risk assessment in the pre-construction stage, without considering on-site safety upon actual execution which shall involve video analytics. Ho et al. [34] injected textual prompts with an image segmentation model to identify the floor elevation from street view images. Chen et al. [15] developed a visual construction safety query chatbot that combines real-time image captioning, safety-centric VQA and keyword-based image-text retrieval. Sun et al. [35] introduced a specialized VLM aimed at recognizing construction waste, integrating a pre-feature interaction network to better facilitate prompt engineering.

The existing studies in the literature show that a standardized prompting framework for construction safety monitoring is lacking. Current VLMs lack contextual awareness and suffer from hallucination in complex tasks or unfamiliar scenarios, when applied to a specific domain without proper prompting. Existing studies of prompt engineering in the construction industry mainly catered for case-by-case response to short queries, while lacking a comprehensive prompt design to align with the construction domain knowledge (e.g. a full list of safety regulations). A knowledge-enhanced prompting framework will guide a VLM to analyze and report unsafe worker behaviors more adaptively.

2.3. Ontology for construction safety analysis

Ontology is structured representations of knowledge that consist of entities, attributes, and relationships between them [36,37], enabling machines to reason and infer new knowledge. Ontologies have been

adopted for several civil engineering applications, including building information modeling [38–40], automated building design performance optimization [41–43] and construction supply chain management [44,45]. For construction safety, several studies [46–49] defined semantic ontology to represent safety-related knowledge (e.g. key entities and their hierarchical relationships) but have not integrated with on-site image analytics for automated compliance checking. Shen et al. [50] defined an ontology as the reasoning basis for identifying safety risks by analyzing the planned construction process/schedule of a building. Xing et al. [51] also developed an ontology for inferring safety risks from construction project statements in planning stage. However, these studies only focused on design for safety but have not integrated with real-time on-site condition monitoring.

As for construction site monitoring, construction safety ontology encapsulates domain-specific information regarding safety regulations, hazards and best safety practices, which can facilitate construction safety compliance checking. Several studies integrated CV models with NLP algorithms for on-site safety compliance checking [52,53], utilizing well-established ontology for more structured and comprehensive analysis of safety conditions. Zhang et al. [48] utilized Mask-RCNN to detect individual objects and Transformer to infer object pair-wise relationship. Then, they combined BERT with a keyword mapping mechanism to integrate domain knowledge from regulation, finally to generate hazard description. However, their categorization of rule violation was limited in scope, only checking if workers wear PPE or not based on straightforward object detection. Analyses of the more complex worker motion for behavioral monitoring are needed. Fang et al. [53] defined an ontology to facilitate spatial proximity/conflict analysis among workers, PPE and moving equipment. Pan et al. [54] proposed a knowledge extraction framework to extract entities from videos/images and perform semantic mapping with a construction activity ontology. However, these studies still utilized ontology as an external module to align the objects extracted from images frames with safety rules. The process of handling vision and language modules separately rather than in an end-to-end model is inefficient and time-consuming, due to the inherent complexity of multi-modal data correlation and feature aggregation. Moreover, a critical issue arises during the transfer of safety information across different modalities: features that are not extracted in the vision modality may be lost when represented in the language modality. This loss of information can lead to incomplete, inaccurate or even distorted representations of the safety conditions, potentially compromising the effectiveness of the compliance checking process.

To address the above limitations, the potential integration of construction safety ontology with VLM prompt design may enhance the multimodal feature extraction in an end-to-end manner, which is still under-explored. More crucially, the ontology models used in conventional methods require a sophisticated system design and frequent updates to keep track of the ever-changing conditions on the construction site. This constant need for updates can pose significant challenges in terms of maintenance cost and scalability. Meanwhile, the in-context learning prompt of a VLM can be updated more efficiently via simple edition in natural language. The integration of ontology in an in-context prompt can address the challenges in both cost and time.

3. Proposed methodology

To address the fundamental challenges in adapting VLMs for construction safety applications, a comprehensive framework is proposed to systematically tackle the data scarcity issue and the domain adaptation difficulty. The overall methodology is illustrated in Fig. 1, mainly including: (1) Literature review for defining a list of safety rules for compliance checking; (2) Semi-automatic data generation with an in-context learning pipeline; (3) Development of a context-aware VLM with a two-stage curriculum learning paradigm; (4) Design of a domain-tailored prompting framework driven by a construction safety ontology. (5) Deployment on a real construction site for real-time video analytics

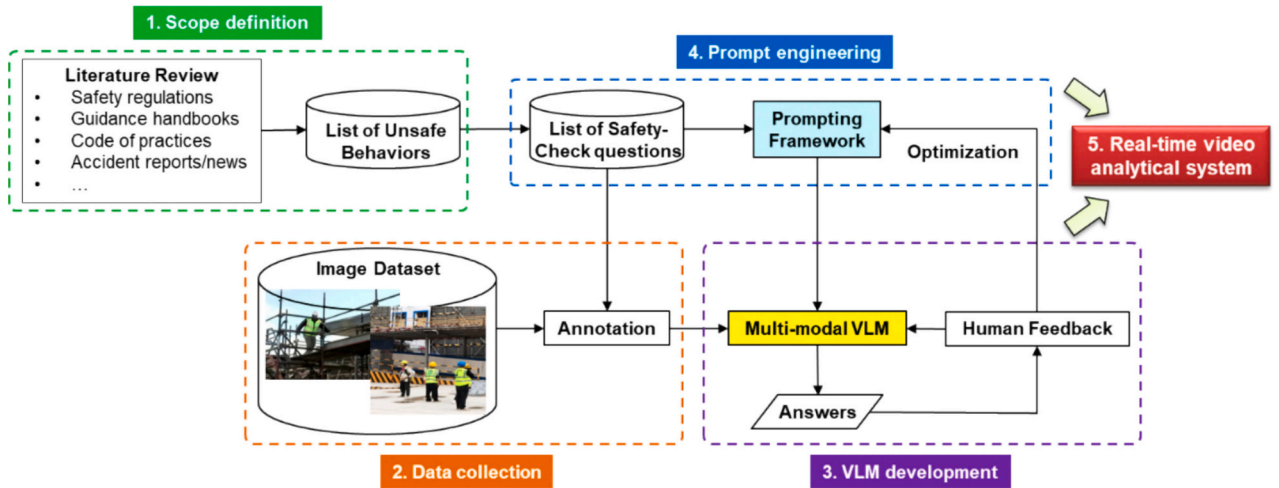


Fig. 1. Proposed methodology for developing a VLM-based safety monitoring system.

with incident reporting and interactive VQA.

3.1. Semi-automatic image-text data labeling pipeline

This section describes a streamlined workflow to generate a high-quality, multi-modal and instruction-following dataset with key-object

coordinates for developing a domain-tailored VLM toward construction safety compliance monitoring. A semi-automatic and self-instructed annotation pipeline driven by an in-context learning framework is designed, as illustrated in Fig. 2.

The data generation pipeline utilizes the pre-trained capability of GPT-4 V and GroundingDINO, i.e., open-set object recognition of

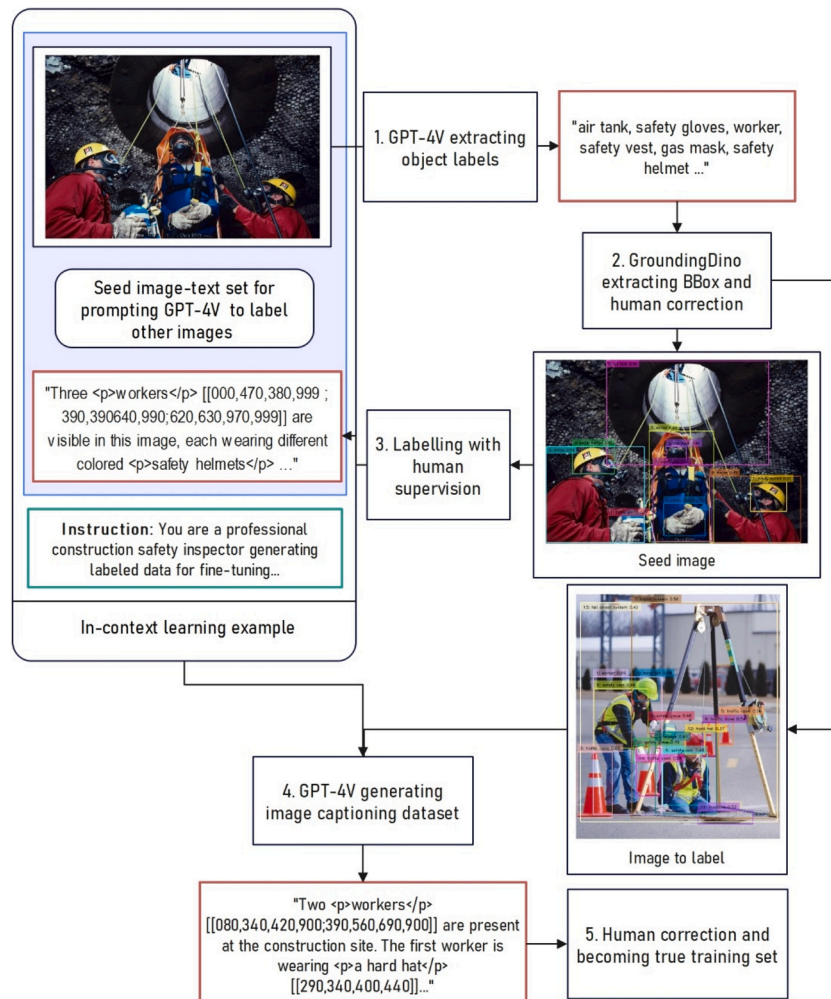


Fig. 2. Semi-automatic self-instructed data labeling pipeline based on in-context learning.

GroundingDINO as well as multimodal understanding and prompting of GPT-4 V. Specifically, the semi-automatic labeling pipeline performs the following five steps:

- 1) **Automatic Label and Bounding Box Generation:** GPT-4 V is utilized to generate a key object list (pseudo labels) of each image in batches. The pseudo labels are then passed to GroundingDINO as text prompts, in order to generate candidate bounding boxes for the specified kinds of objects.
- 2) **Pseudo Label Correction:** While the combination of pre-trained GPT-4 V and GroundingDINO demonstrates competitive zero-shot image processing accuracy, their inferred pseudo-labels and bounding boxes are manually inspected to ensure accuracy and reliability.
- 3) **Seed Image Selection and Hand-Crafting:** A seed image is chosen from the annotated images of each safety scenario (e.g. confined space as shown in Fig. 3), and the description of the seed image is hand-crafted according to the verified bounding boxes either as grounded captioning (detailed descriptions) or question answering of safety compliance analysis. This forms an example of the instruction-response dataset that we want to acquire.
- 4) **In-context Learning for Dataset Generation:** The seed image with the example instruction-response set is passed to an in-context learning prompt, for GPT-4 V to generate candidate instruction-response sets for remaining images. The in-context learning prompt designed for acquiring instruction-response sets for grounded captioning in this paper is summarized below:

System Message: You are a professional construction safety inspector generating labeled data for fine-tuning a safety domain-specific Vision Large Language Model.

You will be given the accurate data of bounding boxes of some workers and objects in the given annotated image. The coordinates of the BBoxes are in 'x1,y1,x2,y2' format with the origin at the top left corner, the x-axis to the right, and the y-axis downward, (x1,y1) and (x2,y2) as top-left and bottom-right corners of the box respectively and values in relative coordinates multiplied by 1000 (prefixed with zeros to three digits).

Please help me draft an instruction for a brief description of the safety condition shown in the image and the corresponding response, and the response should include reference to the given bounding boxes of workers and objects. Please count the number of workers and give a number ID to each worker (ignore the BBox IDs of the workers and start the counting

and ID numbering from 1, from left to right of the image). Also, include each worker's BBox and any of his appearance (like vest color or helmet color) when you mention his safety condition. The given BBox data is sorted in ascending order of x1 coordinate (from left to right). Please include request for grounding with coordinates in the instruction part, and keep the response concise within 200 words, step-by-step and specific to the content of the image. Only include things that you are sure about. Now you will be given an example for reference:

Example:

{

Image: # An image of a construction site with a few workers and a tower crane.

Objects: [

{“value”: 1, “label”: “worker”, “box”: “140,320,240,720”};

... # List of candidate objects with corresponding label and bounding box.

{“value”: 8, “label”: “red safety helmet”, “box”: “814,294,890,360”}.

]

Instruction: Please describe the work environment and safety condition in the image, and include the worker IDs (count from left to right) and coordinates for crucial objects in format [object] [[x0,y0,x1,y1]].

Response: The image depicts a construction site with a tower crane in the background and five workers visible:

- [Worker 1] [[000,330,100,740]] is not wearing a safety helmet or vest.

- [Worker 2] [[140,320,240,720]] has a white [safety helmet] [[164,325,202,368]], but lacks a safety vest.

- [Worker 3] [[370,230,520,980]] is wearing a red [safety helmet] [[410,230,490,300]], yet lacks a safety vest.

- [Worker 4] [[480,260,600,910]] is without any safety helmet or vest.

- [Worker 5] [[800,290,970,999]] sports a red [safety helmet] [[814,294,890,360]], but lacks a safety vest.

Immediate action is required to ensure all workers are equipped with safety helmets and high-visibility vests.

}

You need to rephrase the instruction in a natural way. In the response, the grounded objects with BBoxes should be marked in format [object] followed by coordinates in format [[x1,y1,x2,y2]]. Use the format [[x1,y1,x2,y2;x1,y1,x2,y2;x1,y1,x2,y2;...]] (No spacing between sets of coordinates) for multiple BBoxes of the same object type. You should also pay attention to whether each worker wears sufficient PPE, especially a safety helmet and a safety vest. Now, you will be given the image and key objects for your output

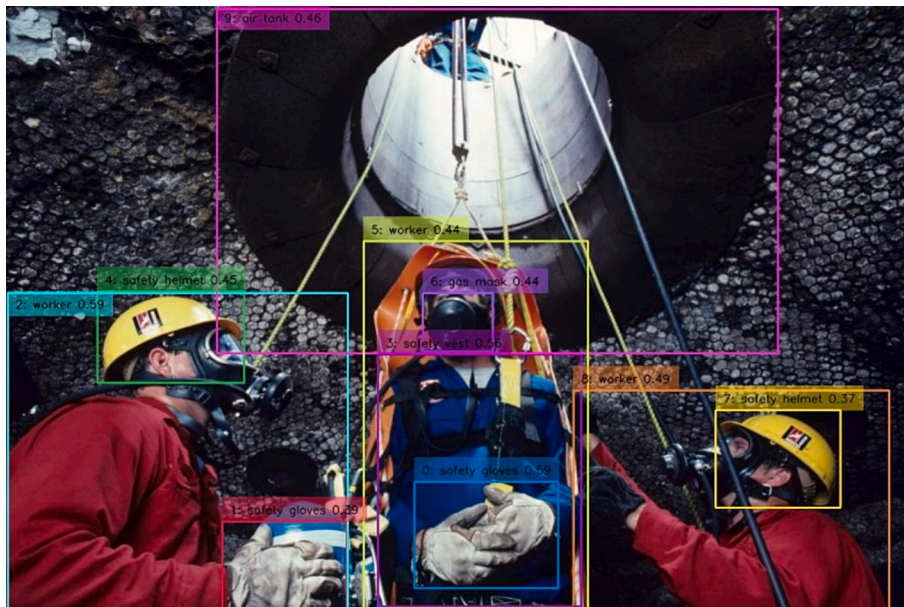


Fig. 3. Seed image annotated by GPT-4 V + GroundingDINO.



Fig. 4. Sample output of semi-automatic data labeling pipeline.

of the corresponding instruction and response:

Image: #The image to be used in the dataset.

Objects: #The key object list of the image with the IDs and coordinates.

Output:

The candidate instruction-response sets generated by the above prompt may contain certain faults and require further polishing.

5) *Polishing Candidate Datasets*: The candidate instruction-response sets generated by GPT-4 V are further optimized with human supervision to create more desired samples in the dataset, each forms an image-instruction-response triplet. A sample output is illustrated in Fig. 4, following the conversational format adopted by VLM-related research [13]. The instruction across different images is slightly different to inject textual diversity, which fosters the VLM to learn more robust and generalizable features from the input instruction.

3.2. Two-stage curriculum learning paradigm for a context-aware VLM

With the image-instruction-response triplet dataset generated from 3.1, a context-aware VLM is then developed for high-level reasoning among the multi-modal data. Previous studies reported that VLMs pre-trained with generic dataset suffer from hallucination, i.e. inaccurate responses due to misinterpreted knowledge for a specific domain. To more effectively integrate domain knowledge into a VLM and to address the aforementioned data imbalance problem, this paper proposes a two-stage curriculum learning paradigm to enhance VLM fine-tuning.

Specifically, the architecture of CogAgent [55] is chosen as our baseline model, with added-on LoRA adapters used for fine-tuning, as illustrated in Fig. 5. CogAgent is built on a pre-trained VLM, specifically the CogVLM 17B, which is an open-source and state-of-the-art large vision language model.

EVA2-CLIP-E is used as the encoder for low-resolution image input (490×490 pixels), and an MLP adapter to map its output into the feature space of the visual-language decoder. The visual-language decoder consists of both vision experts and language backbone for effective deep fusion between image feature and language feature. The decoder processes a combined input of the low-resolution image feature sequence and text feature sequence, and autoregressively outputs the target text. However, the original CogVLM can only accommodate images of relatively low resolution (e.g. 224 or 490), which is insufficient for screen resolution of computers or smart devices which is typically 1280×720 pixels or higher. To address this, a high-resolution but size-efficient cross-module is introduced in the CogAgent architecture, which not only maintains efficiency with high-resolution images but also offers flexible adaptability to a variety of visual-language model architectures. The high-resolution cross-module acts as a new branch for higher-resolution input, accepting images of size 1120×1120 pixels. Unlike the original low-resolution input branch, the high-resolution cross-module adopts a much smaller pre-trained vision encoder (EVA2-CLIP-L with 0.30B parameters), and uses cross-attention of a small hidden size to fuse high-resolution image features with every layer of the VLM decoder, thus reducing the computational cost. The formulation of each decoder layer's attention module is summarized in Eqs. (1)–(2):

$$X'_i = \text{MSA}(\text{layernorm}(X_{in}^i)) + X_{in}^i \quad (1)$$

$$X_{out}^i = \text{MCA}(\text{layernorm}(X'_i), X_{hi}) + X'_i \quad (2)$$

MSA and MCA respectively represent multi-head self-attention of the decoder visual expert and the multi-head cross-attention between the high-resolution module and the decoder. The cross-attention with high-resolution images can be perceived as a complement to the features of low-resolution images, thereby effectively utilizing the previous pre-trained model in low resolution. CogAgent exhibits exceptional performance on benchmarks that assess referring expression comprehension (REC), such as RefCOCO, RefCOCO+, and RefCOCOg, whose grounding capability is comparable to GroundingDINO.

Current studies adopted a staged learning scheme to train VLMs more effectively. For instance, MiniGPT-v2 was trained with three stages (pretraining, multi-task training and multi-modal instruction tuning) for better visual-textual feature alignment [16], with the data decreasing in volume but increasing in human supervision across stages. Inspired by such principle, a two-stage curriculum learning paradigm is proposed to equip our VLM with domain knowledge and capabilities in multiple safety-related tasks, as illustrated in Fig. 6.

The two-stage curriculum learning paradigm mimics the learning process of humans, which starts with easier tasks and ends up with more challenging tasks to help the model learn domain knowledge more efficiently:

1) *Stage 1 – Domain Feature Adaptation*: The first stage involves fine-tuning of the VLM on a set of construction site images with grounded captioning, for injecting construction-specific domain knowledge given the objects of interest annotated with labels, coordinates and inter-relationships. The bounding box annotation can increase the model's attention toward relevant image pixels during this learning stage, helping it align with domain cross-modal knowledge more efficiently. This stage serves to convert the pre-trained VLM into a construction-aware VLM, which excels in related descriptive tasks. As descriptive-task data and crowdsourced online images are more readily prepared and available, we mainly include this kind of data for fine-tuning in this stage.

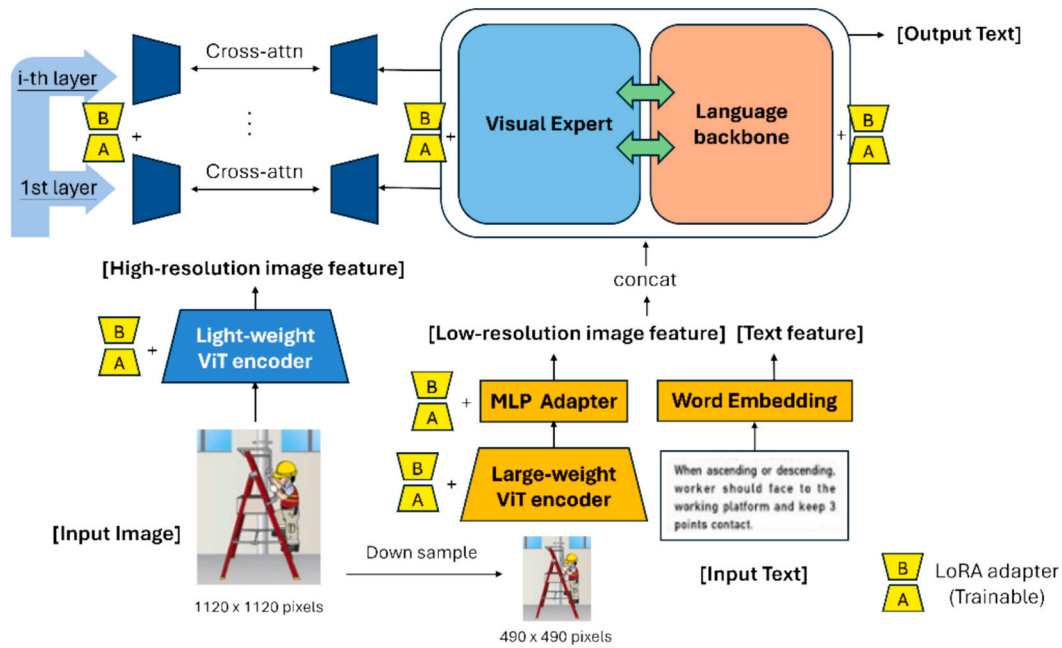


Fig. 5. Architecture of our proposed VLM architecture (revised upon Hong et al. 2024).

Stage 1: Domain Feature Adaptation



For *descriptive* tasks

From *pre-trained* to *construction-aware*

Stage 2: Safety-oriented Instruction Tuning

What is happening on the site?

For domain-tailored *reasoning*

A worker is working near an opening. He is wearing a green hard hat.

Any safety risk or violation?

From *construction-aware* to *safety-oriented*

The floor edge is not protected by guardrails and toe boards, which are compulsory for working at height.

Fig. 6. Illustration of two-stage curriculum learning paradigm.

2) *Stage 2 – Safety-oriented Instruction Tuning*: The second stage focuses on tuning the instruction-following mechanism of the VLM on domain tasks, toward site-specific safety rules to gain task-specialized performance in safety compliance inspection. This stage serves to transform the construction-aware VLM (after finishing Stage 1) into a safety-oriented virtual assistant with context-aware reasoning capability. By learning from more specific safety compliance/violation cases and their analysis in the samples, the model is encouraged to perform logical thinking based on the specific safety requirements. As these kinds of samples require more annotation about sophisticated reasoning and more real-site image for reference, they are less readily available. Therefore, the dataset is less in volume but more supervised in this stage, which corresponds to the common paradigm in conventional VLM fine-tuning and effectively solves the data imbalance problem.

The before-and-after analysis in Fig. 7 demonstrates the enriched and more domain-tailored reasoning capability of our VLM after fine-tuning.

When prompted by an image with workers working at height, the pre-trained VLM (a generalist) only considers a general safety rule of wearing helmet, and does not provide useful recommendations to safety supervisors. After injecting construction domain knowledge, our context-aware VLM (a specialist) is capable of analyzing more complex scenarios with more actionable recommendations, e.g. unguarded opening with a notable risk of falling, hence guardrails and personal fall arrest equipment are suggested.

3.3. Hierarchical prompting framework based on construction safety ontology

Subsequently, the fine-tuned CogAgent is used for VQA process with different work-at-height scenario images. During the inference, an effective VLM prompting framework has been developed to further enhance the VQA performance of CogAgent. Prompting refers to providing an initial input or context to VLM to generate a desired response or continuation. The prompts can be in the form of a question, sentences, code snippets or examples. The prompts serve as a starting point or a specific instruction to guide VLM when generating responses, ensuring that it aligns with the desired context or topic. It provides initial context and constraints that guarantee reasonable and domain-oriented answers output by VLM. Several prompting principles are proposed as general guidelines for different applications, including:

- **Chain-of-Thought (COT) Prompting [31]**: The COT principle enables LLMs to tackle complex arithmetic or reasoning tasks by multiple-step intermediate reasoning and get the correct final answer from steps. This method is particularly useful for tasks that require logical reasoning or problem-solving skills.
- **Least-to-Most (LTM) Prompting [56]**: The LTM principle is a method that solves a complex problem by breaking it down into a series of simpler sub-problems. The process consists of two stages: Decomposition and Sub-problem solving. In the Decomposition stage, instruction or few-shot examples that demonstrate the decomposition are provided in the prompt, followed by the specific question to be decomposed. In the Sub-problem solving stage, few-shot examples demonstrating how subproblems are solved can be provided, then the language model is queried to sequentially solve the subproblems to obtain final answers. LLM with least-to-most

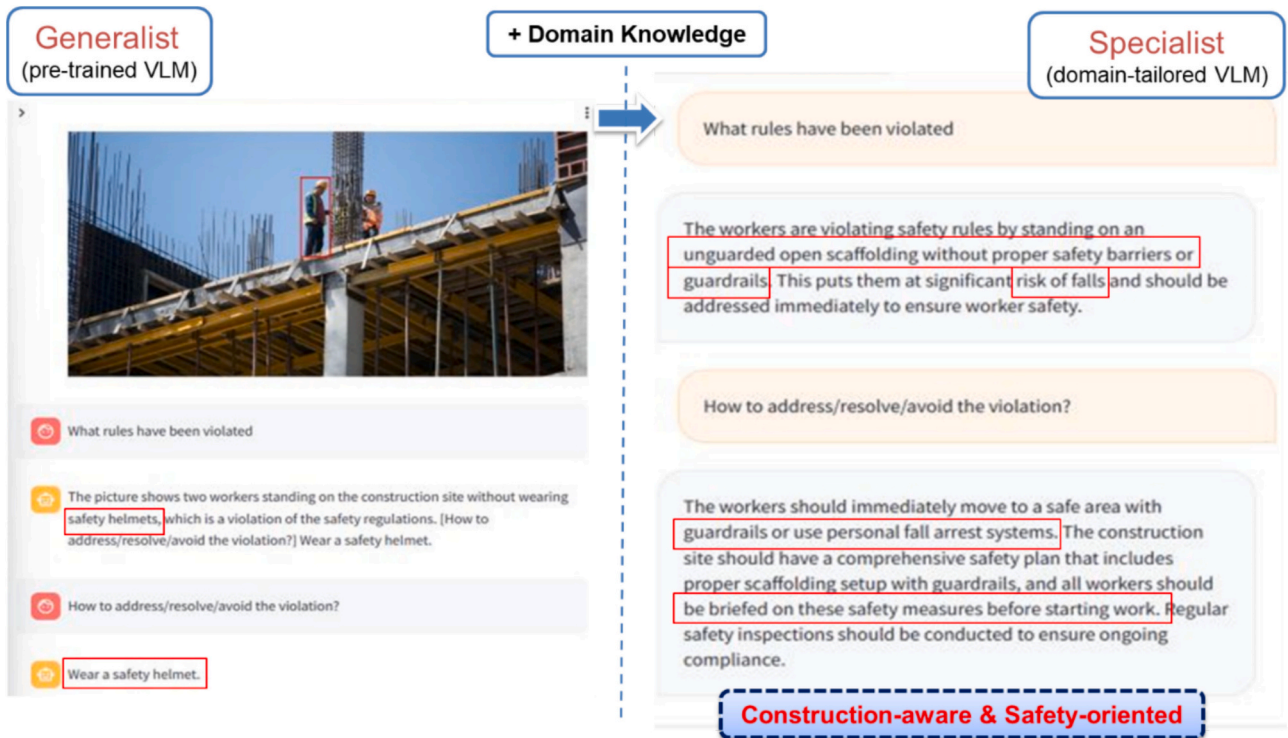


Fig. 7. Illustration of knowledge-enhanced VLM's improved reasoning capability.

prompting is capable of generalizing to more difficult problems than those seen in the prompts or training sets.

During the inference, an effective prompting framework is developed to further enhance the VQA performance of VLM. The motivation is to embed construction-related knowledge into CogAgent as part of the prompts. The list of safety rules/regulations/questions are summarized and then enriched by several strategies:

- (1) *Domain-specific role definition*: A specific role is defined for the VLM in second-person perspective ("As a professional construction site safety supervisor..."), increasing its likelihood to perform more domain-suited reasoning and generate answers that involve more suitable keywords (e.g. PPE compliance, safety risks, etc.).
- (2) *Structured responses*: The format/structure of each intermediate response of our VLM is well-defined (e.g. prompting it to output JSON, with a list of pre-defined key-value pairs). Such data

structuring ensures more deterministic output at each step, allowing the VLM to effectively interpret the intermediate output from previous step as the basis for performing subsequent step.

- (3) *Prompt chaining*: By combining the COT and LTM techniques, the complex problem of safety compliance checking is broken down into easier sub-problems, forming a systematic procedure to analyze an image step-by-step.

Since the original safety rules tend to be abstract/generic for VLM to interpret and perform reasoning, the original descriptions are first converted into a set of prompts that are more suitable for the VQA process of our VLM engine based on construction ontology. Fig. 8 summarizes our proposed hierarchical workflow for generating a construction safety ontology.

The ontology generation framework consists of four levels:

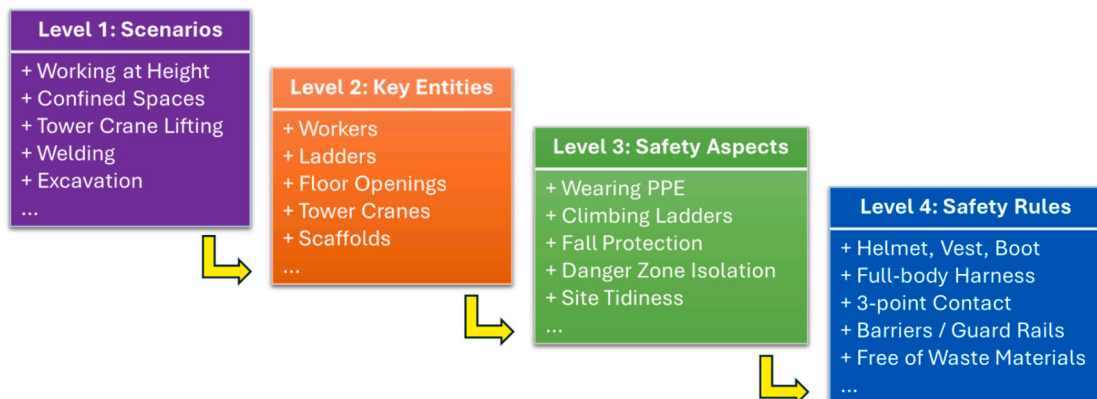


Fig. 8. Proposed hierarchical framework for generating a construction safety ontology.

- 1) *Level 1 – Scenarios*: The specific construction task(s) involved in the monitoring process should be identified for clearly defining the scope of safety requirements, e.g. working at height, lifting or excavation.
- 2) *Level 2 – Key entities*: Based on the list of safety rules defined, the essential objects/equipment/machines relevant to the safety analyses are identified, e.g. ladders/scaffolds for working at height, tower cranes for lifting, etc.
- 3) *Level 3 – Safety aspects*: Based on the entities identified, the categories of possible unsafe behaviors are identified, e.g. PPE is needed if workers are concerned, ladder climbing should be monitored if involving work-at-height tasks, danger zone should be defined if tower crane lifting is necessary, etc.
- 4) *Level 4 – Safety rules*: Based on the defined scope (scenarios, entities and safety aspects), a specific list of safety rules is then defined, which should be categorized into the relevant safety aspects. Such grouping will benefit the prompting framework design subsequently.

It is noted that this workflow is generic, which can be adapted to different site-specific safety requirements. For a more comprehensive illustration in this paper, a list of essential work-at-height safety rules are extracted from the Safety Manual established by Drainage Service Department (2018), and the Work-at-Height Safety Handbook co-published by Development Bureau and Construction Industry Council of Hong Kong (2019). These handbooks also provide a detailed explanation of each safety rule, with graphical illustration that notably facilitated our understanding of how to fine-tune the VLM for monitoring the behaviors and reasoning about the rule analysis. The list of work-at-height behaviors are:

- a. Powered-Operated Elevating Working Platforms (PEWP)
 - i. Wear full body safety harness with its lanyard anchored to a specified anchorage point
- b. Metal Scaffolds
 - i. When erecting, altering dismantling of scaffolds or it is impracticable to erect a safe working platform or provide safe access and egress, the use of full body safety harness attached to secure anchorage point or an independent lifeline is required
- c. Light-Duty Working Platform/ Mobile Working Platform
 - i. When ascending or descending, worker should face to the working platform and keep 3 points contact
 - ii. Only three types of platforms are allowed to carry out work-at-height tasks, including hop-up platform, step platform and mobile platform
 - iii. The surrounding of working platforms should be kept free from waste and miscellaneous materials
- d. Floor Opening/Edge Protection
 - i. Provide guard-rails and toe-boards at the floor edge
 - ii. Provide secure coverings with warning signs at the floor opening.
 - iii. Provide guard-rails, toe boards and warning signs at the floor opening

- iv. Whist installing, alternating or dismantling fall protection facilities at the floor edge, opening and windows, suitable fall arresting system should be provided to workers

With the scope defined for the work-at-height regulations listed above, a guideline for key objects and safety aspects to inspect is formed for safety monitoring. A construction safety ontology is generated correspondingly, as shown in Fig. 9, which logically represents the semantic and hierarchical relationships among the concepts/entities/behaviors related to working at height.

Based on the generated construction safety ontology, a domain-tailored prompting framework is further designed:

- i. Identify the relevant scenario(s) from an image, based on specific objects/features present in the image. This forces the VLM to focus on checking against the relevant safety rules. In particular, the VLM is prompted to first *detect whether each of the following objects exist*. The output is a JSON with the following keys, each with a value either *True/False* to denote the presence of the corresponding objects.
 - a. Worker
 - b. Light-duty mobile platform
 - c. Power-elevated working platform (PEWP)
 - d. Floor opening
- ii. The JSON output in Step(i) is further processed. For each of the objects marked with *True*, the VLM is prompted to check the compliance against those relevant safety rules associated with that object:
 - a. Light-duty mobile platform
 - i. Approved type of platform (step platform vs ladder)
 - ii. Waste or miscellaneous materials around the platform
 - b. Light-duty mobile platform & Worker
 - i. Facing to the platform
 - ii. Maintaining three-point contact
 - c. PEWP & Worker
 - i. Safety harness
 - ii. Secured to anchorage point
 - d. Floor opening
 - i. Fully covered by safety net
 - ii. Guardrails, toe boards and warning signs along the edge
- iii. Report specifically the safety rules that are violated after the analysis in Step(ii), with a brief description of the violation

3.4. Real-time video analytical system

The VLM-based safety compliance monitoring system operates in two different modes: (1) *automatic reporting mode* and (2) *interactive chatbot mode*.

In the first mode, the real-time video from each camera is streamed continuously to the computer, where video frames are sampled at regular intervals to be fed into the VLM for inference (10 s in our field

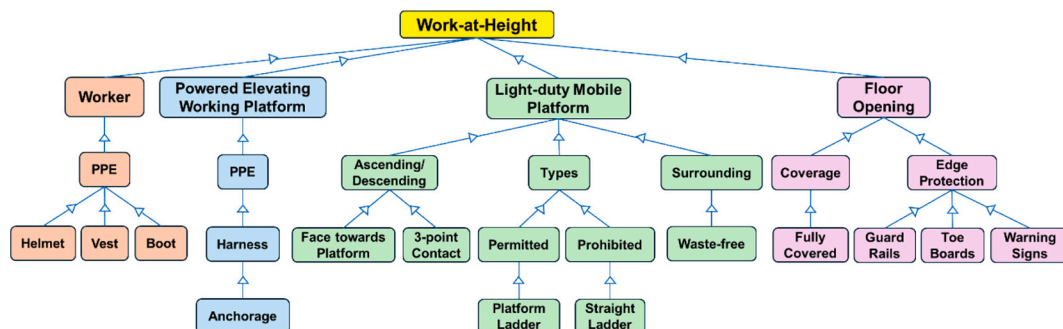


Fig. 9. Proposed construction safety ontology based on the hierarchical framework.

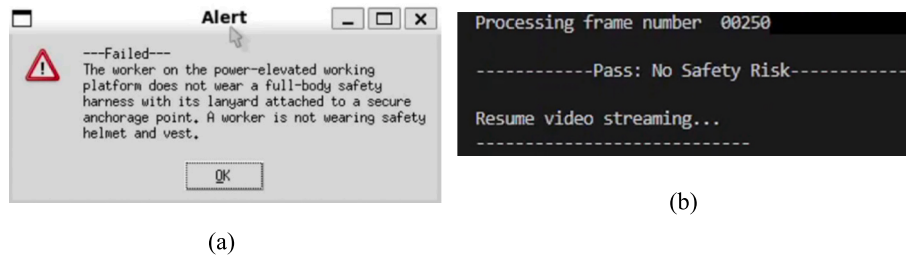


Fig. 10. Illustrations of (a) descriptive alert message and (b) no-alert frame processing output generated by our alarm system.

trials). Then, the VLM automatically generates appropriate answers to the image, based on our three-step prompting framework designed in Section 3.3. On the other hand, the second mode can be user-triggered via a simple button to extract the current frame at any time. Based on the frame, site engineers or safety supervisors can input any question to prompt the VLM to interactively generate personalized answers.

An alarm mechanism is developed to generate descriptive alert messages to safety supervisors. As illustrated in Fig. 10, in case any safety rule violation is identified, such “Failed” safety compliance is reported via an alarm from the onsite computer, where the safety supervisors are notified in real-time. A descriptive alert message is then prompted out, concisely summarizing which specific safety rules are violated. In case no violation is observed from an image, as illustrated in Fig. 10, it passes the safety compliance checking by our VLM which results in a simple message “No Safety Risk” output without triggering the alarm system.

In addition, an incident logging system is developed to store all the processed images and generated output. These data are properly indexed/labeled, which constitute a well-managed database for record-keeping, which also allows safety supervisors to query historical incidents and search for associated images for verification. As illustrated in Fig. 10, each processed frame is indexed with a number (e.g. “00250” denotes the 250th frame of a video stream), which serves as the key for future queries. All these indexed frames are stored, each of which is then labeled with the final output generated by the VLM (i.e. the texts denoting any safety rules violated), which serve as the value for future queries.

4. Validation and discussion

The developed VLM system was deployed at a real construction site in the Shek Wu Hui Sewage Treatment Works (SWHSTW). This site is a secondary sewage treatment plant occupying 9.4 ha of land and handling 81,000 m³ of sewage per day produced by a population of 300,000 in Sheung Shui and Fanling Districts.

4.1. Experimental setup

The dataset for finetuning CogAgent model consists of 1500 construction site images, after applying augmentation techniques including random cropping, horizontal flipping and color space alternation. The images were then annotated with the semi-automatic pipeline proposed in Section 3.1. As shown in Fig. 11, the raw images were first processed by the data labeling pipeline to obtain precise bounding boxes with diverse textual description and instruction-following samples.

The textual descriptions are based on the list of safety questions summarized. Some features/objects in the images are highly relevant to certain safety rules (e.g. scaffolds, working platforms), thus are specifically highlighted in the annotations to ensure that such knowledge can be incorporated into our VLM engine. The incorporated examples also aim to teach CogAgent about the domain COT and LTM prompting techniques designed, which decompose safety analyses into step-by-step question-answering to obtain final reasonable analysis (i.e. by first

determining the safety scenario that image belongs to and then going on analyzing the corresponding safety rules).

The VLM is then fine-tuned with our two-stage curriculum learning paradigm proposed in Section 3.2. The data split is 6:4, i.e. 60 % (900 images) for training and validation, the remaining 40 % (600 images) for testing. Among the training and validation sets, 600 were collected from the internet via keywords search on objects and scenes related to the work-at-height scope, and the remaining 300 were collected from the target site. For the testing set, all 600 images were collected from the target site. The fine-tuning was done with a single NVIDIA A40 GPU with 48 GB of available VRAM. For the training configuration, due to the limited amount of computational resources and dataset, PEFT is rather suitable for supervised fine-tuning than full-parameter training [27]. LoRA is adopted to achieve PEFT, with all the original weights frozen. Five LoRA modules, each with a rank of 64, and a LoRA alpha value of 128, were injected into the visual language backbone, the high-resolution cross-attention layers, the Vision Transformer encoder and the MLP adapter respectively. The VLM was trained with a batch size of 2 for 2 epochs for each of the two stages, with a warmup ratio of 0.1, an initial learning rate of 3e-5, and a cosine learning rate scheduler. The LoRA dropout was set to 0.1 and the weight decay was set to 0.05. After finetuning, the VLM was quantized to 8 bit and deployed in one RTX-4090 GPU with 24 GB of VRAM.

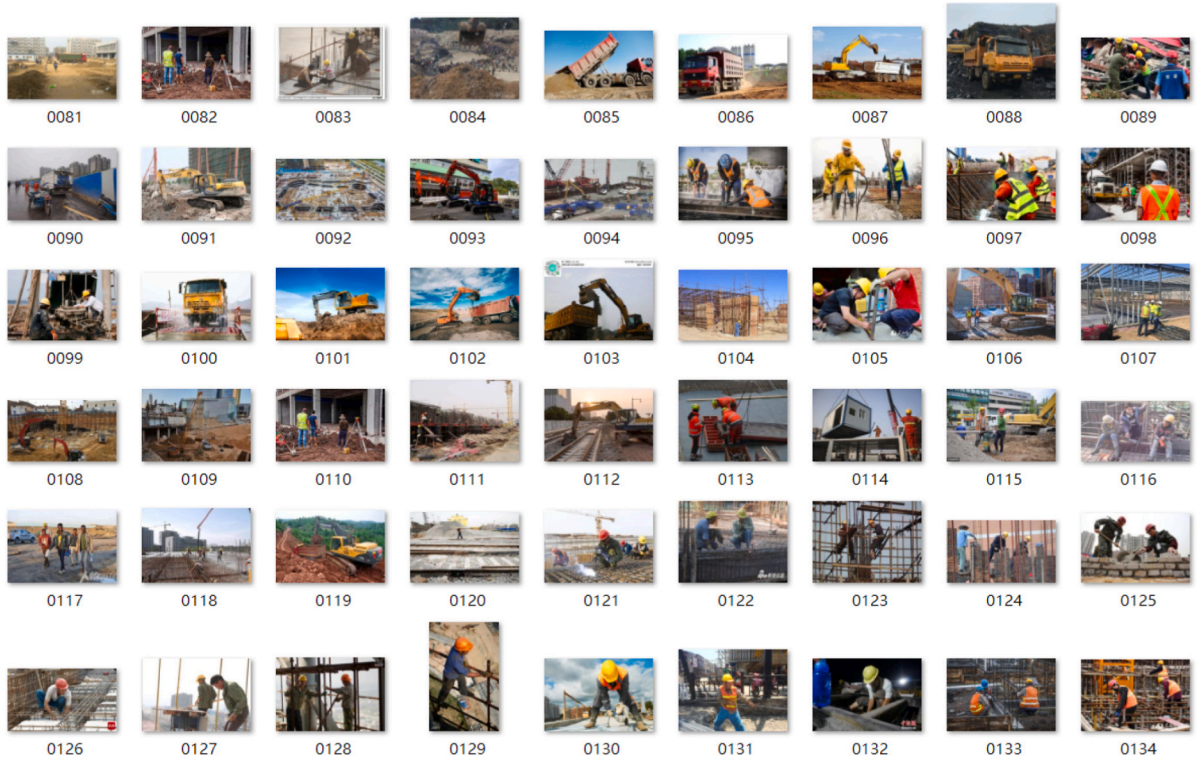
The fine-tuned VLM was then integrated with our ontology-driven prompting framework designed in Section 3.3, as well as our real-time video analytical system developed in Section 3.4. The entire monitoring system was finally deployed to the SWHSTW construction site, with four cameras covering the major work-at-height areas on two floors of an industrial building.

4.2. Quantitative evaluation

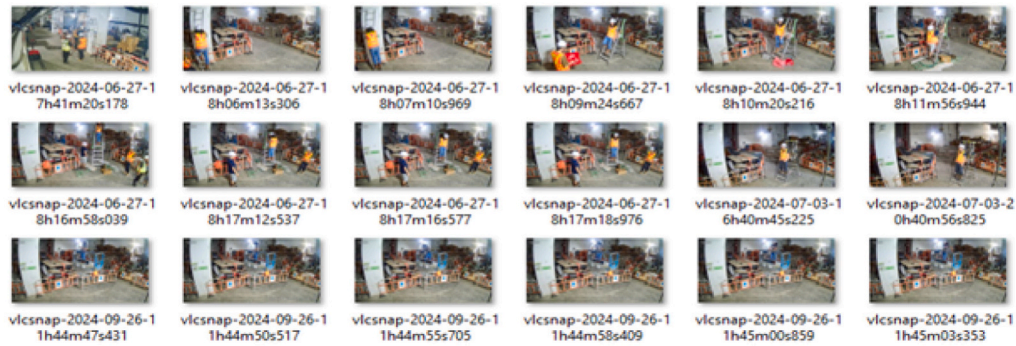
The images collected from the real site were then fed into our VLM to carry out VQA. The performance of safety compliance checking was evaluated with the protocol summarized in Table 1. In the context of safety compliance checking in this project, two possible outcomes were defined: (1) **Compliance – Pass**, where no safety rule is violated, and (2) **Compliance – Failed**, where a particular safety rule is violated. Therefore:

- True Positive (TP) means that an image does contain violation scenario(s), and the VLM correctly identifies them. Note that the evaluation is counted on a per-rule basis rather than a per-image basis.
- False Negative (FN) means that an image does contain violation scenario(s), but the VLM cannot identify them and wrongly output “Pass – No Safety Risk”.
- False Positive (FP) means that an image has no safety violation, but the VLM wrongly outputs “Failed” for a particular safety rule.
- True Negative (TN) means that an image has no safety violation, and the VLM correctly outputs “Pass – No Safety Risk”.

These measures are the foundation for our two primary evaluation metrics, *Sensitivity* and *Specificity*, defined in Eqs. (3)–(4):



(a)



(b)

Fig. 11. Snapshot of the collected (a) crowdsourced and (b) real-site image datasets for VLM fine-tuning and testing.

$$\text{Sensitivity (\%)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Specificity (\%)} = \frac{TN}{TN + FP} \quad (4)$$

- *Sensitivity* denotes the percentage of violation scenarios correctly identified by VLM.
- *Specificity* denotes the percentage of compliant scenarios correctly passed without alert.

In the context of construction safety monitoring, these metrics carry

Table 1

Proposed evaluation metrics for safety compliance checking.*

Ground-truth \ Output	Compliance – Failed	Compliance – Pass
Compliance – Failed	TP	FN
Compliance – Pass	FP	TN

* The values denote the number of safety rules being checked one-by-one.

significant practical implications. Sensitivity measures the system's ability to correctly identify actual safety violations, such as missing PPE or unsafe behaviors at heights. This metric is particularly crucial as missing violations (false negatives) could lead to serious accidents or

Table 2
Quantitative results among different methods.

Variant	Strategy 1: Ontology-driven prompting	Strategy 2: Two-stage curriculum learning	Sensitivity	Specificity
Baseline	×	×	76.3 %	74.3 %
Strategy 1	✓	×	86.5 %	85.5 %
Strategy 2	×	✓	84.7 %	91.7 %
Strategy 1 + 2	✓	✓	90.3 %	92.5 %

fatalities. Complementarily, specificity measures the system's accuracy in correctly identifying safe conditions while avoiding false alarms. High specificity is essential for maintaining operational efficiency, as false positives can cause unnecessary interruptions on the site engineers' daily routines, possibly diminishing their confidence in the monitoring system.

The quantitative results among the implementation of different methods are summarized in Table 2. Specifically, four variants were compared, including the baseline one without our proposed strategies, and the remaining three variants incorporating different combinations of the two main strategies, i.e., ontology-driven prompting and two-stage curriculum learning, proposed in this paper.

For the baseline variant, the original model without fine-tuning is used for inference, with no special prompting strategy. All rules were directly fed into the VLM at once, asking the VLM in a straightforward manner to output the ones being violated. With such direct prompting, the baseline model suffered from information overflow and insufficient reasoning capability, leading to suboptimal sensitivity and specificity rates of 76.3 % and 74.3 % respectively. This phenomenon suggested that the model struggled to distinguish pertinent details amidst the vast data inputs.

Hence, our ontology-driven prompting strategy effectively structured the model's reasoning process by leveraging hierarchical and logical thinking based on the work-at-height safety knowledge. This method substantially boosted sensitivity to 86.5 % and specificity to 85.5 %, indicating a more effective retrieval of relevant information. Still, the pre-trained vision backbone module might have limited the VLM in discerning finer domain-specific details.

Hence, further improvements were achieved by our two-stage curriculum learning strategy, which effectively adapted the pre-trained VLM to construction domain and work-at-height scenarios, significantly enhancing the specificity to 91.7 %. This increase indicated a more refined ability to identify accurate information. However, a slight decrease in sensitivity to 84.7 % was observed, possibly due to the persistent challenge of information overflow, which required prompt engineering to deal with.

The most notable enhancement was achieved by combining both strategies proposed in this paper (i.e. two-stage curriculum learning paradigm and ontology-driven prompting framework), which harmoniously addressed the weaknesses of both individual methods. This dual approach leveraged the strengths of domain-tailored model adaptation and the context-aware reasoning process. The result was a remarkable achievement, with 91.5 % sensitivity and 90.5 % specificity, demonstrating the efficacy of this integrated approach for construction safety monitoring. The comparative analysis shows that both proposed methods can effectively improve the performance of the system in terms of both accuracy and operation efficiency.

4.3. Qualitative evaluation

Besides quantitative evaluation, qualitative evaluation is also an intuitive way to inspect the performance of our domain-enhanced VLM. Based on the work-at-height safety rules listed in Section 3.3, the output generated by our VLM (for compliance "Pass" and "Failed" respectively)

are illustrated below.

- Case 1: Light-duty Mobile Platform
- "Failed": The results are shown in Fig. 12 (a) where a ladder was correctly identified as an non-approved type of mobile platform, and Fig. 12 (b) where the waste or miscellaneous materials around the ladder were successfully detected. Both correctly identified incidents had triggered the alarm system to prompt out alert messages with a brief description of the violation. It is noted that a worker was around the ladder but not ascending/descending it. There was also a PEWP next to the ladder and worker but not being used at that moment. Hence, only the two relevant safety rules (mobile platform type and surrounding materials) were checked, and our VLM is capable of filtering out other irrelevant rules during inference.
- "Pass": The results are shown in Fig. 12 (c) and Fig. 12 (d). Our VLM correctly identified the mobile platform being used as a step platform (one of the approved types), and its surrounding was free of waste or miscellaneous materials. Under such scenario without violating any safety rule, the output "Pass: No Safety Risk" was printed only on screen for safety supervisors to verify the VLM's response to the input image.
- Case 2: Light-duty Mobile Platform (with Worker Climbing)
- "Pass": As shown in Fig. 13 (a), the worker ascending/descending the step platform was facing directly to the platform and maintaining a three-point contact with it. Hence, the safety compliance checking resulted in a "Pass".
- "Failed": As shown in Fig. 13 (b), the worker was facing away from the step platform, and his hands were not holding the step platform safely. In this case, the alarm system was triggered with an alert message prompted out, stating the safety rules being violated.
- Case 3: PEWP (with Worker in Operation)
- "Pass": As shown in Fig. 14 (a), the worker standing on top of the PEWP was wearing safety harness, with its lanyard attached to a secure anchorage point. The VLM simply printed "Pass: No Safety Risk" on screen to verify his safety compliance, without triggering any alarm.
- "Failed": As shown in Fig. 14 (b), the worker on the PEWP was not wearing a safety harness. Our VLM correctly identified such safety violation and generated the corresponding alert message.
- Case 4: Floor Opening/Edge
- "Pass": As shown in Fig. 15 (a), the floor opening was fully covered by safety net, with guardrails and toe boards protecting the floor edge. It is noted that a warning sign was not visible in the image, while the protective measures for that floor opening already sufficed. Hence, the VLM eventually outputted a "Pass".
- "Failed": As shown in Fig. 15 (b), the upper portion of the floor opening was not covered by safety net, leaving a noticeable gap that pose falling hazard. In addition, there was no warning sign around the floor edge. Under such scenario, our VLM correctly identified these safety risks and generated the alert messages "not fully covered" and "no warning sign".

Overall, our VLM demonstrated competitive performance in work-at-height safety compliance monitoring. However, two cases were found particularly challenging: (1) For identifying a three-point contact with the mobile platform, the worker's limbs are frequently occluded when climbing the mobile platform (e.g. when facing to the right, his left hand and left leg are occluded respectively by his body and right leg); (2) For determining whether a safety harness is attached to a secure anchorage point (e.g. a PEWP), the thinness of the rope and small size of the anchor make the object identification difficult. More crucially, the anchor is usually occluded by the worker or the PEWP itself, further challenging the detection of anchorage. These are the edge cases where compliance

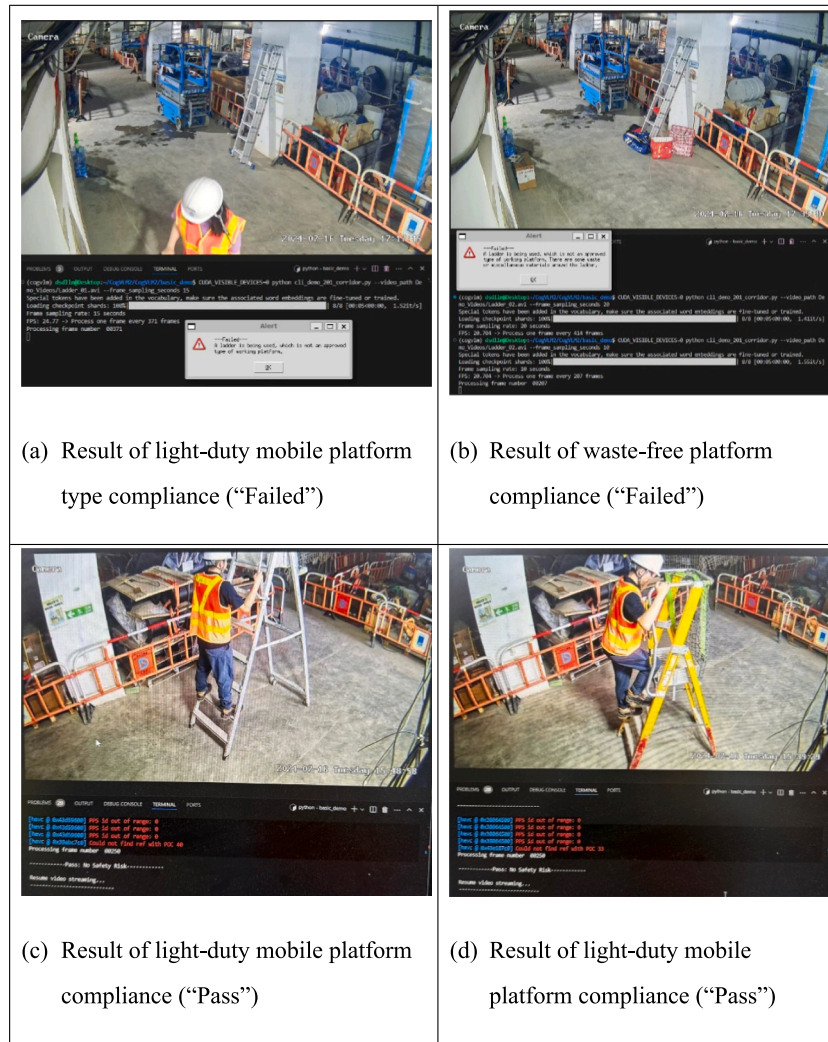


Fig. 12. Results of compliance checking for Light-duty Mobile Platform.

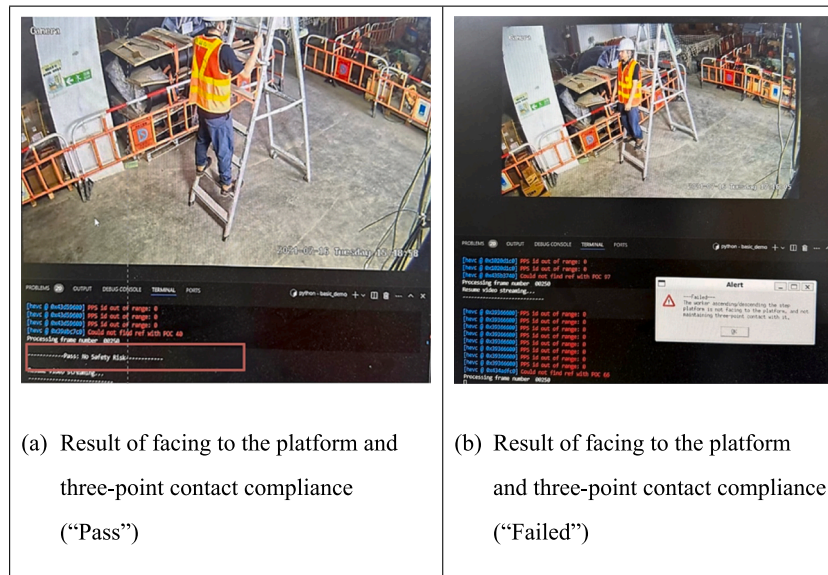


Fig. 13. Result of facing to the platform and three-point contact compliance for Light-duty Mobile Platform.

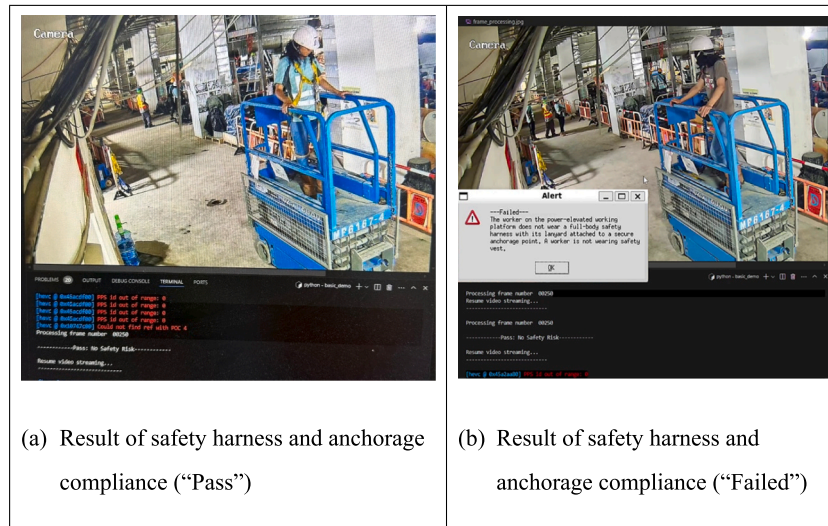


Fig. 14. Result of safety harness and anchorage compliance for PEWP with worker in operation.

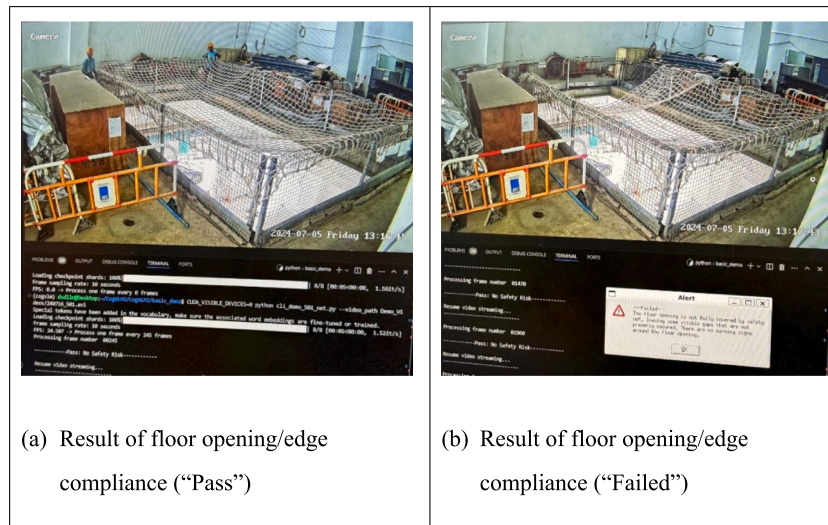


Fig. 15. Result of compliance for floor opening/edge protection.

checking accuracy will be compromised. Future research shall develop more specialized techniques to ensure robustness of the analytical system in such cases.

5. Conclusion and future work

This paper addressed critical challenges in adapting VLMs for construction safety monitoring through a comprehensive framework. Our semi-automatic self-instructed pipeline, combining GPT-4 V and GroundingDINO for instruction-following dataset generation, successfully overcomes the data scarcity problem. The two-stage curriculum learning also addresses the data imbalance between online images and real-site images. With our curriculum learning paradigm and ontology-driven prompting framework, the VLM-based monitoring system demonstrates remarkable effectiveness in domain knowledge integration. Upon deployment on a real construction site in Hong Kong, the integrated system achieved a sensitivity of 90.3 % and a specificity of 92.5 % in work-at-height safety compliance checking, substantially improved over the baseline method.

These research findings have significant implications for both academia and industry. Our framework provides a scalable solution for

adapting VLM-based systems to specialized domains with limited data availability, while the success demonstrated in real-world deployment establishes a practical pathway for automating construction safety monitoring. Compared with existing conventional CV-NLP safety monitoring systems which require significant data, human labor and sophisticated system design upon development, our innovative system can be developed and set up more efficiently, and allows for deep multimodal feature extraction and a higher level of reasoning as demonstrated in this paper. The simple interface design also allows the safety officer, whether on site or in remote position, to interact with the VLM easily and greatly facilitates site safety monitoring. The high accuracy achieved in real-site safety compliance inspection suggests significant potential for widespread adoption in construction site management, enabling more proactive and efficient safety protocols.

At the same time, there also exist some edge cases where the system's robustness may be compromised due to limitation in visual information processing, e.g., small object detection and occlusion issues. More sophisticated strategies, such as multi-frame temporal correlation, can be achieved by fine-tuning with multi-frame data or in-context learning framework with more comprehensive prompting for processing multiple image frames. Multiple frames are more reliable than a single frame

when processing complicated visual information such as action recognition or scenes with occlusion. For small object detection, an external CNN-based module may be incorporated into our framework for detection and localization of small objects, to enhance the robustness in visual information processing without compromising efficiency (i.e., CNN-based models are computational-efficient when compared to VLMs).

Another limitation lies in the ontology-driven approach, such as requiring update of the ontology-based prompt to accommodate new safety regulations or site-specific variations. When compared with conventional methods, ours can already allow for more efficient adaptation of the ontology by editing in natural language rather than machine language. Nonetheless, this process can be further automated through ontology extraction and prompt conversion upon various site-specific regulations by a LLM specialist. Automated approaches for ontology adaptation can be made feasible by LLM prompt engineering, as LLM specializes in analyzing long pieces of regulation and extracting key objects and relationships for building ontology. Future research will focus on enhancing the system through the above vision processing methods and ontology-update method for improving contextual awareness and scalability. The framework's generalizability can also be further validated across a more diverse set of safety rules and a variety of site conditions, and be extended to analytics of more complex worker or machine actions for more comprehensive safety monitoring practices.

CRedit authorship contribution statement

Chak-Fu Chan: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Peter Kok-Yiu Wong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Xiaowen Guo:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Jack C.P. Cheng:** Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Jolly Pui-Ching Chan:** Resources, Project administration, Investigation, Data curation. **Pak-Him Leung:** Software, Resources, Project administration, Funding acquisition, Data curation. **Xingyu Tao:** Writing – review & editing, Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to acknowledge the project (No. DEMP/2023/25) funded by Drainage Services Department, the Government of the Hong Kong Special Administrative Region, Wanchai, Hong Kong Island, Hong Kong, SAR, for providing support to this research. The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Labour Department of, H.G, Occupational Safety and Health Statistics 2018. https://www.labour.gov.hk/eng/osh/pdf/archive/statistics/OSH_Statistics_2018_eng.pdf, 2018 (last accessed May 24, 2025).
- [2] Labour Department of, H.G, Occupational Safety and Health Statistics 2019. https://www.labour.gov.hk/eng/osh/pdf/archive/statistics/OSH_Statistics_2019_eng.pdf, 2019 (last accessed May 24, 2025).
- [3] U.S. Bureau of Labor Statistics, Census of Fatal Occupational Injuries Summary 2023. <https://www.bls.gov/news.release/cfoi.nr0.htm>, 2023 (last accessed May 24, 2025).
- [4] M. Wang, P. Wong, H. Luo, S. Kumar, V. Delhi, J. Cheng, Predicting safety hazards among construction workers and equipment using computer vision and deep learning techniques, in: ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction, IAARC Publications, 2019, <https://doi.org/10.22260/isarc2019/0054>.
- [5] J.C.P. Cheng, P.K.Y. Wong, H. Luo, M. Wang, P.H. Leung, Vision-based monitoring of site safety compliance based on worker re-identification and personal protective equipment classification, *Autom. Constr.* 139 (2022) 104312, <https://doi.org/10.1016/j.autcon.2022.104312>.
- [6] H. Wu, J. Zhao, An intelligent vision-based approach for helmet identification for work safety, *Comput. Ind.* 100 (2018) 267–277, <https://doi.org/10.1016/j.compind.2018.03.037>.
- [7] H. Luo, M. Wang, P.K.Y. Wong, J. Tang, J.C.P. Cheng, Construction machine pose prediction considering historical motions and activity attributes using gated recurrent unit (GRU), *Autom. Constr.* 121 (2021) 103444, <https://doi.org/10.1016/j.autcon.2020.103444>.
- [8] H. Luo, M. Wang, P.K.Y. Wong, J.C. Cheng, Full body pose estimation of construction equipment using computer vision and deep learning techniques, *Autom. Constr.* 110 (2020) 103016, <https://doi.org/10.1016/j.autcon.2019.103016>.
- [9] J. Tang, M. Wang, H. Luo, P.K.Y. Wong, X. Zhang, W. Chen, J.C. Cheng, Full-body pose estimation for excavators based on data fusion of multiple onboard sensors, *Autom. Constr.* 147 (2023) 104694, <https://doi.org/10.1016/j.autcon.2022.104694>.
- [10] N. Khan, M.R. Saleem, D. Lee, M.W. Park, C. Park, Utilizing safety rule correlation for mobile scaffolds monitoring leveraging deep convolution neural networks, *Comput. Ind.* 129 (2021) 103448, <https://doi.org/10.1016/j.compind.2021.103448>.
- [11] S. Paneru, I. Jeelani, Computer vision applications in construction current state, opportunities & challenges, *Autom. Constr.* 132 (2021) 103940, <https://doi.org/10.1016/j.autcon.2021.103940>.
- [12] Y. Wang, B. Xiao, A. Bouferguene, M. Al-Hussein, H. Li, Vision-based method for semantic information extraction in construction by integrating deep learning object detection and image captioning, *Adv. Eng. Inform.* 53 (2022) 101699, <https://doi.org/10.1016/j.aei.2022.101699>.
- [13] OpenAI, GPT-4V(ision) System Card. https://cdn.openai.com/papers/GPTV_System_Card.pdf, 2023 (last accessed May 24, 2025).
- [14] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu, L. Zhang, Grounding dino Marrying dino with grounded pre-training for open-set object detection, *arXiv* (2023), <https://doi.org/10.48550/arXiv.2303.05499> preprint arXiv:2303.05499.
- [15] H. Chen, L. Hou, S. Wu, G. Zhang, Y. Zou, S. Moon, M. Bhuiyan, Augmented reality, deep learning and vision-language query system for construction worker safety, *Autom. Constr.* 157 (2024) 105158, <https://doi.org/10.1016/j.autcon.2023.105158>.
- [16] J. Chen, D. Zhu, X. Shen, X. Li, Z. Liu, P. Zhang, R. Krishnamoorthi, V. Chandra, Y. Xiong, M. Elhoseiny, Minigpt-v2 large language model as a unified interface for vision-language multi-task learning, *arXiv* (2023), <https://doi.org/10.48550/arXiv.2310.09478> preprint arXiv:2310.09478.
- [17] Y.C. Zhou, Z. Zheng, J.R. Lin, X.Z. Lu, Integrating NLP and context-free grammar for complex rule interpretation towards automated compliance checking, *Comput. Ind.* 142 (2022) 103746, <https://doi.org/10.1016/j.compind.2022.103746>.
- [18] B. Xiao, Y. Wang, Y. Zhang, C. Chen, A. Darko, Automated daily report generation from construction videos using ChatGPT and computer vision, *Autom. Constr.* 168 (2024) 105874, <https://doi.org/10.1016/j.autcon.2024.105874>.
- [19] W. Wang, Q. Lv, W. Yu, W. Hong, J. Qi, Y. Wang, J. Ji, Z. Yang, L. Zhao, X. Song, J. Xu, CogVLM Visual expert for pretrained language models, *arXiv* (2023), <https://doi.org/10.48550/arXiv.2311.03079> preprint arXiv:2311.03079.
- [20] H. Liu, C. Li, Y. Li, Y.J. Lee, Improved baselines with visual instruction tuning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, <https://doi.org/10.1109/CVPR52733.2024.02484>.
- [21] J. Li, D. Li, S. Savarese, S. Hoi, Blip-2 Bootstrapping language-image pre-training with frozen image encoders and large language models, in: International Conference on Machine Learning, PMLR, 2023, <https://doi.org/10.48550/arXiv.2301.12597>.
- [22] X. Chen, X. Wang, S. Changpinyo, A.J. Piergiovanni, P. Padlewski, D. Salz, S. Goodman, A. Grycner, B. Mustafa, L. Beyer, A. Kolesnikov, Pali A jointly-scaled multilingual language-image model, *arXiv* (2022), <https://doi.org/10.48550/arXiv.2209.06794> preprint arXiv:2209.06794.
- [23] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou, J. Zhou, Qwen-vl A frontier large vision-language model with versatile abilities, *arXiv* (2023), <https://doi.org/10.48550/arXiv.2308.12966> preprint arXiv:2308.12966.
- [24] C. Fan, Q. Mei, X. Li, Assisting in the identification of ergonomic risks for workers a large vision-language model approach, in: ISARC Proceedings of the International Symposium on Automation and Robotics in Construction, IAARC Publications, 2024, <https://doi.org/10.22260/isarc2024/0131>.
- [25] Y. Jung, I. Cho, S.H. Hsu, M. Golparvar-Fard, VisualSiteDiary a detector-free vision-language transformer model for captioning photologs for daily construction reporting and image retrievals, *Autom. Constr.* 165 (2024) 105483, <https://doi.org/10.1016/j.autcon.2024.105483>.
- [26] Z. Han, C. Gao, J. Liu, S.Q. Zhang, Parameter-efficient fine-tuning for large models: a comprehensive survey, *arXiv* (2024), <https://doi.org/10.48550/arXiv.2403.14608> preprint arXiv:2403.14608.

- [27] E.J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, Lora Low-rank adaptation of large language models, arXiv (2021), <https://doi.org/10.48550/arXiv.2106.09685> preprint arXiv:2106.09685.
- [28] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, Language Models are Unsupervised Multitask Learners. https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf, 2019 (last accessed May 24, 2025).
- [29] Q. Dong, L. Li, D. Dai, C. Zheng, J. Ma, R. Li, H. Xia, J. Xu, Z. Wu, T. Liu, A survey on in-context learning, arXiv (2022), <https://doi.org/10.48550/arXiv.2301.00234> preprint arXiv:2301.00234.
- [30] O. Rubin, J. Herzig, J. Berant, Learning to retrieve prompts for in-context learning, arXiv (2021), <https://doi.org/10.48550/arXiv.2112.08633> preprint arXiv:2112.08633.
- [31] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q.V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, Adv. Neural Inf. Proc. Syst. 35 (2022) 24824–24837, <https://doi.org/10.48550/arXiv.2201.11903>.
- [32] G. Yong, K. Jeon, D. Gil, G. Lee, Prompt engineering for zero-shot and few-shot defect detection and classification using a visual-language pretrained model, Comput. Aided Civ. Inf. Eng. 38 (11) (2023) 1536–1554, <https://doi.org/10.1111/mice.12954>.
- [33] S. Wong, C. Zheng, X. Su, Y. Tang, Construction contract risk identification based on knowledge-augmented language models, Comput. Ind. 157 (2024) 104082, <https://doi.org/10.1016/j.compind.2024.104082>.
- [34] Y.H. Ho, L. Li, A. Mostafavi, Integrated vision language and foundation model for automated estimation of building lowest floor elevation, Comput. Aided Civ. Inf. Eng. 40 (1) (2025) 75–90, <https://doi.org/10.1111/mice.13310>.
- [35] Y. Sun, Z. Gu, S.B. Yang, Probing vision and language models for construction waste material recognition, Autom. Constr. 166 (2024) 105629, <https://doi.org/10.1016/j.autcon.2024.105629>.
- [36] S. Ji, S. Pan, E. Cambria, P. Marttinen, S.Y. Philip, A survey on knowledge graphs representation, acquisition, and applications, IEEE Trans. Neur. Netw. Learn. Syst. 33 (2) (2022) 494–514, <https://doi.org/10.1109/TNNLS.2021.3070843>.
- [37] C. Ramonell, R. Chacón, H. Posada, Knowledge graph-based data integration system for digital twins of built assets, Autom. Constr. 156 (2023) 105109, <https://doi.org/10.1016/j.autcon.2023.105109>.
- [38] W. Chen, K. Chen, J.C.P. Cheng, Towards an ontology-based approach for information interoperability between BIM and facility management, Adv. Comp. Strateg. Eng. (2018), https://doi.org/10.1007/978-3-319-91638-5_25. Cham: Springer.
- [39] H. Liu, J.C.P. Cheng, V.J. Gan, S. Zhou, A novel data-driven framework based on BIM and knowledge graph for automatic model auditing and quantity take-off, Adv. Eng. Inform. 54 (2022) 101757, <https://doi.org/10.1016/j.aei.2022.101757>.
- [40] H. Liu, J.C.P. Cheng, V.J. Gan, S. Zhou, A knowledge model-based BIM framework for automatic code-compliant quantity take-off, Autom. Constr. 133 (2022) 104024, <https://doi.org/10.1016/j.autcon.2021.104024>.
- [41] Z. Wu, J.C.P. Cheng, Z. Wang, H.H. Kwok, An ontology-based framework for automatic building energy modeling with thermal zoning, Energ. Build. 296 (2023) 113267, <https://doi.org/10.1016/j.enbuild.2023.113267>.
- [42] Z. Wu, Z. Wang, J.C.P. Cheng, H.H. Kwok, A knowledge-informed optimization framework for performance-based generative design of sustainable buildings, Appl. Energy 367 (2024) 123318, <https://doi.org/10.1016/j.apenergy.2024.123318>.
- [43] S. Wu, Q. Shen, Y. Deng, J.C.P. Cheng, Natural-language-based intelligent retrieval engine for BIM object database, Comput. Ind. 108 (2019) 73–88, <https://doi.org/10.1016/j.compind.2019.02.016>.
- [44] J.C.P. Cheng, M. Das, Ontology-based standardized web services for context aware building information exchange and updating, Comp. Civ. Eng. (2011), [https://doi.org/10.1061/41182\(416\)80](https://doi.org/10.1061/41182(416)80).
- [45] M. Das, J.C.P. Cheng, K.H. Law, An ontology-based web service framework for construction supply chain collaboration and management, Eng. Constr. Archit. Manag. 22 (5) (2015) 551–572, <https://doi.org/10.1108/ECAM-07-2014-0089>.
- [46] K. Farhaly, R.K. Soman, W. Collinge, P. Manu, M.H. Mosleh, C. Cheung, Construction safety ontology development and alignment with industry foundation classes (IFC), Electron. J. Inf. Technol. Constr. 27 (2022) 94–108, <https://doi.org/10.36680/j.itcon.2022.005>.
- [47] Y. Lu, Q. Li, Z. Zhou, Y. Deng, Ontology-based knowledge modeling for automated construction safety checking, Saf. Sci. 79 (2015) 11–18, <https://doi.org/10.1016/j.ssci.2015.05.008>.
- [48] L. Zhang, J. Wang, Y. Wang, H. Sun, X. Zhao, Automatic construction site hazard identification integrating construction scene graphs with BERT based domain knowledge, Autom. Constr. 142 (2022) 104535, <https://doi.org/10.1016/j.autcon.2022.104535>.
- [49] F. Simone, S.M. Ansaldi, P. Agnello, R. Patriarca, Industrial safety management in the digital era constructing a knowledge graph from near misses, Comput. Ind. 146 (2023) 103849, <https://doi.org/10.1016/j.compind.2022.103849>.
- [50] Q. Shen, S. Wu, Y. Deng, H. Deng, J.C.P. Cheng, BIM-based dynamic construction safety rule checking using ontology and natural language processing, Buildings 12 (5) (2022) 564, <https://doi.org/10.3390/buildings12050564>.
- [51] X. Xing, B. Zhong, H. Luo, H. Li, H. Wu, Ontology for safety risk identification in metro construction, Comput. Ind. 109 (2019) 14–30, <https://doi.org/10.1016/j.compind.2019.04.001>.
- [52] S. Tang, D. Roberts, M. Golparvar-Fard, Human-object interaction recognition for automatic construction site safety inspection, Autom. Constr. 120 (2020) 103356, <https://doi.org/10.1016/j.autcon.2020.103356>.
- [53] W. Fang, L. Ma, P.E. Love, H. Luo, L. Ding, A.O. Zhou, Knowledge graph for identifying hazards on construction sites integrating computer vision with ontology, Autom. Constr. 119 (2020) 103310, <https://doi.org/10.1016/j.autcon.2020.103310>.
- [54] Z. Pan, C. Su, Y. Deng, J.C.P. Cheng, Image2Triplets a computer vision-based explicit relationship extraction framework for updating construction activity knowledge graphs, Comput. Ind. 137 (2022) 103610, <https://doi.org/10.1016/j.compind.2022.103610>.
- [55] W. Hong, W. Wang, Q. Lv, J. Xu, W. Yu, J. Ji, Y. Wang, Z. Wang, Y. Dong, M. Ding, J. Tang, Cogagent a visual language model for gui agents, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, <https://doi.org/10.48550/arXiv.2312.08914>.
- [56] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, D. Schuurmans, C. Cui, O. Bousquet, Q. Le, E. Chi, Least-to-most prompting enables complex reasoning in large language models, arXiv (2022), <https://doi.org/10.48550/arXiv.2205.10625> preprint arXiv:2205.10625.